

Probabilistic Life Expectancy Projections for Puerto Rico Using the Lee-Carter Model

Table of Contents

Abstract

Puerto Rico faces acute demographic challenges, including a rapidly aging population, shrinking tax base, and institutional fiscal stress—that demand robust, forward-looking mortality projections. This thesis applies the Lee-Carter model to Puerto Rico’s age and sex specific mortality data from 1950 to 2023, decomposing historical death rates into their baseline, sensitivity, and trend components. A bootstrapped simulation framework generates 500 probabilistic mortality trajectories, producing 40-year stochastic forecasts of age-specific death rates and life expectancy at birth with 95% confidence intervals. Results project median life expectancy at birth reaching 82 years for males and 94 years for females by 2062, with male projections broadly consistent with those of the UN World Population Prospects, the U.S. Census Bureau’s International Data Base, and the IHME. Female projections, however, exhibit pronounced upward skew that exceeds best-practice benchmarks in the literature, a pattern traced to data irregularities in the 2010–2023 period rather than modeling error. Key structural findings include a plateau in young adult male mortality attributable to excess violence-related deaths and high sensitivity of infant and child mortality to the overall trend index. The wide confidence intervals produced are themselves a substantive finding, signaling that demographic uncertainty in Puerto Rico constitutes a genuine planning risk warranting scenario-based fiscal and healthcare policy rather than single-point forecasts.

Future work should implement a hierarchical Bayesian model and investigate the underlying data quality issues driving the female projection skew.

Introduccion

Background

Mortality projections serve as a foundational pillar in population forecasting, standing alongside fertility and migration as one of the three critical variables that determine the shape of our future society (Preston, 2000). They are essential for understanding future demographic trends and planning for social services, healthcare, and economic development (Kontis et al., 2017). They are a critically important tool for policymakers, demographers, and researchers to anticipate and respond to changes in population dynamics. The failure to accurately project mortality rates can lead to significant misallocations of resources and inadequate planning for future needs.

Within the balancing equation of demography, mortality serves as the primary metric for assessing the health and survival limits of a population. As global societies undergo the epidemiological transition characterized by a shift from infectious diseases to chronic and age-related illness, the ability to forecast future survival patterns has become essential for understanding broader demographic trends (Omram, 2001; Vaupel, 2010). Consequently, mortality projections are no longer merely statistical exercises; they are vital instruments for social planning, economic forecasting, and the development of public health strategies.

For policymakers and researchers, these projections provide the quantitative foundation required to anticipate and respond to the challenges of an aging society. Specifically, accurate forecasts are critical for the solvency of national pension systems and the longterm sustainability of social security frameworks (Bengtsson & Keilman, 2019; R. D. Lee & Carter, 1992). In the area of public health, projections allow for the strategic allocation of resources toward geriatric care and the management of non-communicable diseases that are common in older populations (Christensen et al., 2009). Without robust data, governments risk “longevity risk” the economic threat posed by populations living significantly longer than anticipated by historical models (Cairns et al., 2006).

The historical record demonstrates that failing to accurately project mortality can lead to severe structural consequences. For decades, official forecasts consistently underestimated the pace of mortality decline, particularly at older ages, leading to inadequate planning for healthcare infrastructure and retirement funding (Booth, 2006; Oeppen & Vaupel, 2002). This historical underestimation underscores the necessity of moving beyond simple deterministic models toward modern computerized tools and stochastic frameworks.

Advances in Mortality Modeling

The transition from deterministic to stochastic modeling represents a paradigm shift in the field of demographic forecasting. Historically, mortality projections relied on deterministic methods, such as the Gompertz-Makeham law

of mortality or expert led “scenario based” variants used by institutions like the United States Social Security Administration. These older models often assumed a fixed rate of improvement or relied on subjective “high,” “medium,” and “low” scenarios that failed to capture the non-linear and age-specific nature of mortality decline (Booth, 2006). Critically, deterministic models offer no formal measure of uncertainty; they provide a single trajectory that historically has proven to be consistently over-conservative, leading to the repeated underestimation of life expectancy gains at older ages (R. Lee & Miller, 2001).

The landmark introduction of the Lee-Carter model revolutionized the field by shifting the focus from expert opinion to a data-driven, statistical framework. The Lee-Carter method employs a stochastic approach that decomposes the logarithm of age-specific death rates into three components: an average age profile, a time varying mortality index, and an age-specific response to that index. By using Singular Value Decomposition (SVD) to extract the primary trend of mortality change, the model allows the data itself to dictate the pace of decline rather than imposing arbitrary limits on survival (R. D. Lee & Carter, 1992). This method has since become the “gold standard” in demography (Giroso & King, 2008).

Beyond its predictive accuracy, the primary value of the Lee-Carter framework lies in its stochastic nature. Unlike deterministic models, which offer a “best guess,” stochastic models treat future mortality as a random process with an associated probability distribution. By modeling the time-varying index (k_t) as a stochastic time series typically a random walk with drift, researchers can generate probabilistic intervals for future life expectancy. This quantification of uncertainty is indispens-

able for modern risk management; it allows policymakers and actuaries to calculate “longevity risk” and the probability of extreme outcomes, such as a “95% confidence interval” for the future old-age dependency ratio (Cairns et al., 2006; R. Lee, 2000). In a 21st-century context, where the pace of medical innovation and the impact of global health crises are unpredictable, the ability to communicate uncertainty through these probabilistic frameworks is no longer a luxury, but a prerequisite for responsible fiscal and social planning.

Puerto Rico: Demographic Context and Gap in the Literature

Policymakers and public health officials in Puerto Rico need annual stochastic mortality projections as part of producing annual probabilistic population projections of the island population. The lack of precise, forward looking mortality data has already manifested in severe institutional challenges across Puerto Rico, most notably within its social safety nets and fiscal frameworks. The island’s retirement systems offer a cautionary example of how demographic uncertainty can lead to systemic collapse. For decades, the mismatch between projected and actual dependency ratios, driven by a shrinking tax base and a rapidly aging population, contributed to the total exhaustion of the employees retirement system’s trust funds (Abel & Deitz, 2014; Cox Alomar, 2019). Because policymakers lacked robust, stochastic projections to account for the accelerated pace of aging, the govern-

ment was eventually forced to transition to a “Pay As You Go” or “PayGo” model. This shift has placed an enormous, immediate burden on the central budget, where pension payments now compete directly with essential services like education and public safety (Waldman, 2023). Furthermore, the inability to accurately project mortality trends has complicated the island’s long term recovery and debt restructuring efforts. Strategic documents, such as the certified fiscal plans issued by the Financial Oversight and Management Board, rely heavily on population forecasts to determine Puerto Rico’s future debt paying capacity (Puerto Rico Fiscal Agency and Financial Advisory Authority 2024)

This thesis will use the Lee-Carter model to first analyze the series from 1950 to 2023 into its three components, the a_x , b_x , and k_t parameters by age and sex to obtain a deeper understanding of the underlying mortality trends in Puerto Rico. Then, we will bootstrap the parameters to create a distribution of possible future mortality scenarios, which will allow us to create a stochastic forecast of mortality rates and life expectancy at birth for the next 40 years. Finally, we will compare our projections to those produced by the UN World Population Prospects, the U.S. Census Bureau’s International Data Base, and the University of Washington’s Institute for Health Metrics and Evaluation to contextualize our findings within the broader landscape of demographic forecasting and assess the robustness of the model, identify sources of uncertainty and bias in the projection and provide evidence based recommendations for improving future Puerto Rico specific mortality forecasting efforts

Literary Review

Lee Carter

The Lee-carter model was introduced in 1992 by Ronald Lee and Lawrence Carter in their paper “Modeling and Forecasting U.S. Mortality”(R. D. Lee & Carter, 1992). It revolutionized mortality forecasting by providing a data-driven approach in lieu of the expert-led methods that were previously dominant. Prior to its development, official forecasts often underestimated the gains in human life expectancy (Olshansky, 1988), creating a pressing need for more accurate projections to assess the solvency of pension systems and social security programs.

The model can be described simply with the equation

$$\ln(m_{x,t}) = a_x + b_x k_t + e_{x,t}$$

More verbose, it states that the natural log of the mortality rates of a given year $\ln(m_{x,t})$ are equivalent to the baseline mortality a_x plus the product of the age specific sensitivity to changes in mortality b_x and the time varying mortality index k_t plus an error term $e_{x,t}$. To achieve this, Lee and Carter applied Singular Value Decomposition (SVD) to extract the characteristic pattern of mortality improvement over time. The solution to the SVD is not unique, so they imposed two constraints:

the sum of b_x across ages must equal 1, and the sum of K_t across time must equal 0. This allowed them to identify a unique solution that captured the dominant trend in mortality improvement. Then by only taking the first singular value and its corresponding vectors in a_x and b_x , they reduce the complexity of the data while retaining the most important information. This serves as compression of the data reducing the number of parameters needed to model mortality trends but keeping the essential features of the data. Following this, the time-varying mortality index K_t is modeled as a random walk with drift. A random walk with drift value equals the previous period's value plus a constant drift term and a stochastic error component, producing a process that exhibits both a deterministic and a stochastic trend (Pwasong & Sathasivam, 2015). These allow for the creation of objective prediction intervals that provided a quantifiable measure of uncertainty. This was a significant departure from previous "best-guess" forecasts, which lacked any formal way to assess the range of possible outcomes.

The simplicity and robustness of the Lee-Carter model allowed it to outperform contemporary models (Booth, 2006), closely tracking actual mortality data while older models significantly under-predicted life expectancy gains. Its ability to capture the linear trend of log-mortality decline across the 20th century made it particularly effective for long-term forecasting. Unlike earlier parametric models that focused on the biological shape of the mortality curve at a single point in time, the Lee-Carter model prioritized the dynamic movement of that curve over decades.

Thanks to the nature of the Lee-Carter model extensions and variations have been developed to address its limitations and improve its accuracy. For example,

the LC-ER (Extended Rotation) method was developed to account for “rotation” in mortality decline, which addresses implausible age-pattern artifacts in vanilla LC projections. Additionally, the stmomo R package(Villegas et al., 2018), whose implementation we will be using later, provides a flexible framework for mortality forecasting using the Lee-Carter model as well as several other models, allowing users to fit the model to various data types and generate forecasts while assessing uncertainty.

The model and its extensions have been used in various applications, including forecasting breast cancer mortality in China and Pakistan(Mubarik et al., 2023), where it outperformed simpler models for the screen-age/late-onset population (ages 50–84), revealing rising mortality in Pakistan versus declining rates in China, which was used to inform targeted healthcare interventions. The UN employs the LC model to forecast global mortality trends, especially for long-term projections up to 2100(Wpp, 2022). For high-life-expectancy countries like Japan, the model was extended to account for “rotation” in mortality decline (slowing improvements at young ages and accelerating declines at older ages). This adjustment, termed the LC-ER (Extended Rotation) method, addressed implausible age-pattern artifacts in vanilla LC projections.

Now, when talking about the implementation we will be using, Stochastic Mortality Modeling (stMoMo) is an R package that provides a comprehensive framework for fitting and forecasting mortality models, including the Lee-Carter model. Traditionally, mortality projections followed deterministic frameworks, producing “high,” “medium,” and “low” scenario variants that, while intuitive, offer no coherent

probabilistic interpretation and whose resulting ranges for vital rates and population measures are inherently inconsistent with one another (R. D. Lee & Tuljapurkar, 1994). Recognizing these limitations, the field has progressively shifted toward stochastic approaches, in which uncertainty is explicitly modeled and forecasts are expressed as probability distributions over future outcomes. This transition has been formalized at the institutional level: the 2015 edition of the UN World Population Prospects marked a landmark departure from scenario-based projections toward fully probabilistic ones, reflecting a broader consensus that only stochastic methods can fully and coherently quantify forecasting uncertainty (Alexander & Raftery, 2024) The stMoMo package allows users to easily fit the Lee-Carter model to their mortality data, generate forecasts, and assess uncertainty through probabilistic intervals. It also includes various extensions of the Lee-Carter model, such as the LC-ER method, which can be used to improve the accuracy of forecasts for high-life-expectancy countries. The package is designed to be user-friendly and flexible, making it accessible for researchers and practitioners in the field of mortality forecasting. For our use case, we will be using the base Lee-Carter model and using its initial fit to bootstrap its parameters to then simulate numerous forecasts to create a distribution of possible outcomes. Bootstrapping functions by instead of relying on theoretical assumptions about the distribution of the parameters, it resamples the data with replacement to create new datasets. By fitting the model to these new datasets, we can generate a distribution of parameter estimates that reflects the variability in the data. To do so, we use the link function to set the random component associated to the data. The default link function is the log link, this as-

sumes that the deaths follow a Poisson distribution, which is appropriate for count data like mortality. Secondly, we need to set the restraints for the model to ensure identifiability. For this we will set it as “sum”. This ensures that $\sum_t k_t = 0$. With these two we can now create the seed fit for the bootstrapping process. The bootstrap itself requires its own set of parameters, these being the number of bootstrap samples, the approach to be applied and the type of death to resample.

Methodology

The input matrix

Starting off with the input matrix, this is the matrix of log age specific death rates that we will be using to fit the model. The matrix is organized by age and year, with the log of the death rates for each age group and year. Our original input data for this matrix comes in the form of death counts and population exposures, which we then convert into death rates by dividing the death counts by the population exposures. Finally, we take the log of these death rates to create the input matrix for the Lee-Carter model. The data comes from multiple sources. The population estimates come from the US Census Bureau decennial censuses divided into age groups and sex and interpolated for the intercensal years using a linear spline. The mortality data is a collage of multiple sources (Puerto Rico Department Of Health, WHO database and the Center for Disease Control and Prevention. The data is organized into 5-year age groups and single year of age for the first year of life, and then 85+ for the last group.

We then pass this matrix to the stMoMo package to fit the Lee-Carter model and extract the parameters. After several tests, we determined that 100 samples generated properly distributed residuals, but for robustness, we choose 500 to provide a more robust fit and still iterate at an acceptable pace. Following this we

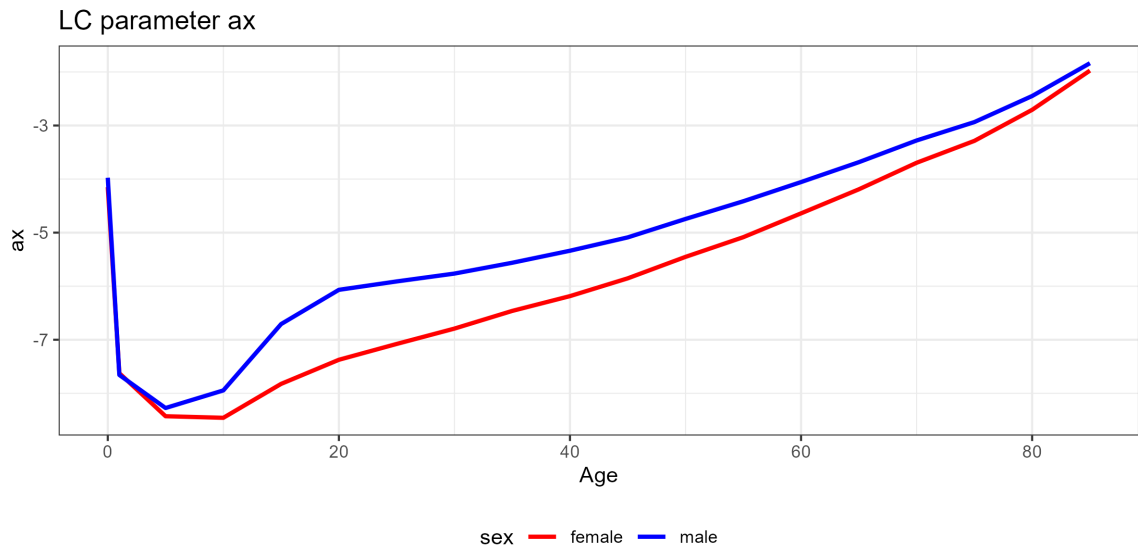
proceed to the simulations. The needed parameters for this are the number of simulations which we will set as the same as the number of bootstraps, the years to project set at 40. The jump choice, whether to use the fitted end or the actual end, we will choose the actual end in the data. Lastly for the simulation is the method of the projection of kt . Our alternatives are multivariate random walk with drift and an N independent arima models. The result of this is a series of 500 by 500 death rates for each of the 40 years to be projected for every age group. We can then select the projections that correspond to the median and the 97.5% and 2.5% quantiles to generate out confidence intervals for the death rates per age group, the first of our final projects. Following this we can turn these into life tables to generate our final product, the life expectancy at birth projections with confidence intervals. The notable life table assumptions we make are available with the code in github, but the main changes are setting the average number of years lived in the first two age groups to empirically derived values based on the infant mortality rate, rather than assuming deaths are uniformly distributed across the interval.

The initial fit

Fitted ax

Once the initial model fitting is completed, we have all our initial products. The first of these, ax parameter, serves as the baseline mortality for the population. It represents the average log mortality rate for each age group across the entire

time period. The ax parameter captures the general shape of the mortality curve, which typically shows high mortality rates at very young ages, a decline in mortality during childhood and early adulthood, and then an increase in mortality as people age. Ours graphs as follows.

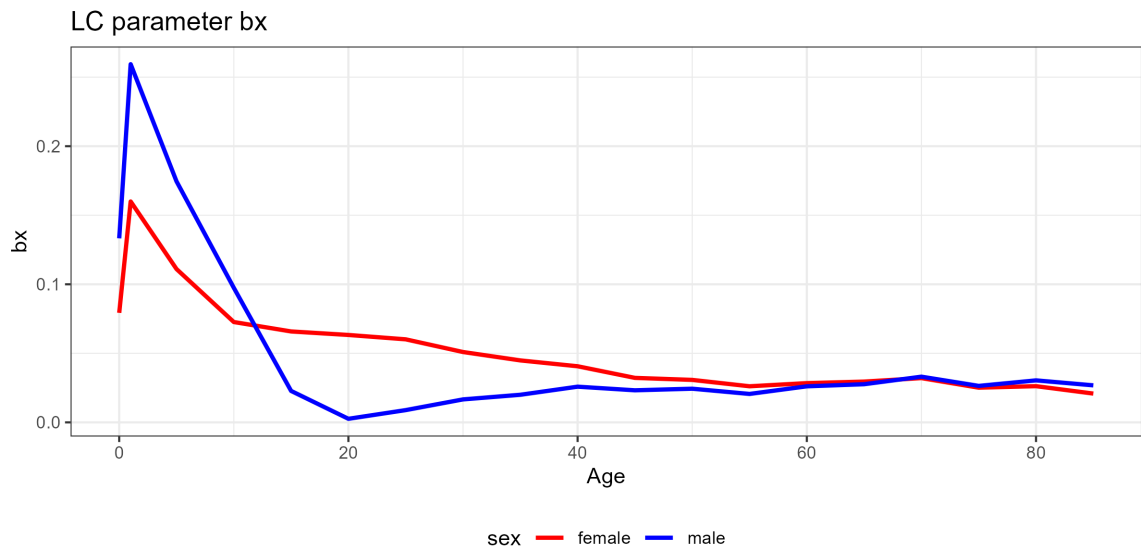


Here, we can see the drastic fall of the baseline log mortality after the first year of life. This is followed by the steady increase as one ages and the telltale male mortality hump starting at the teen years followed by a slow closing of the gap between the sexes as they age.

Fitted bx

Next we have the bx parameter, which is our age specific sensitivity to changes in the mortality index kt . The bx parameter captures how much the log mortality rate for each age group changes in response to changes in the mortality index kt . A higher bx value for a particular age group indicates that the mortality rate

for that age group is more sensitive to changes in the overall mortality trend captured by kt . The bx parameter typically shows a pattern where certain age groups, such as infants and older adults, have higher sensitivity to changes in mortality trends, while other age groups, such as young adults, may have lower sensitivity. This reflects the fact that improvements in healthcare and living conditions can have a more significant impact on reducing mortality rates for certain age groups compared to others. In other words this parameter tends to be context specific and can vary significantly between populations and time periods. For our case, the bx parameter graphs as follows.

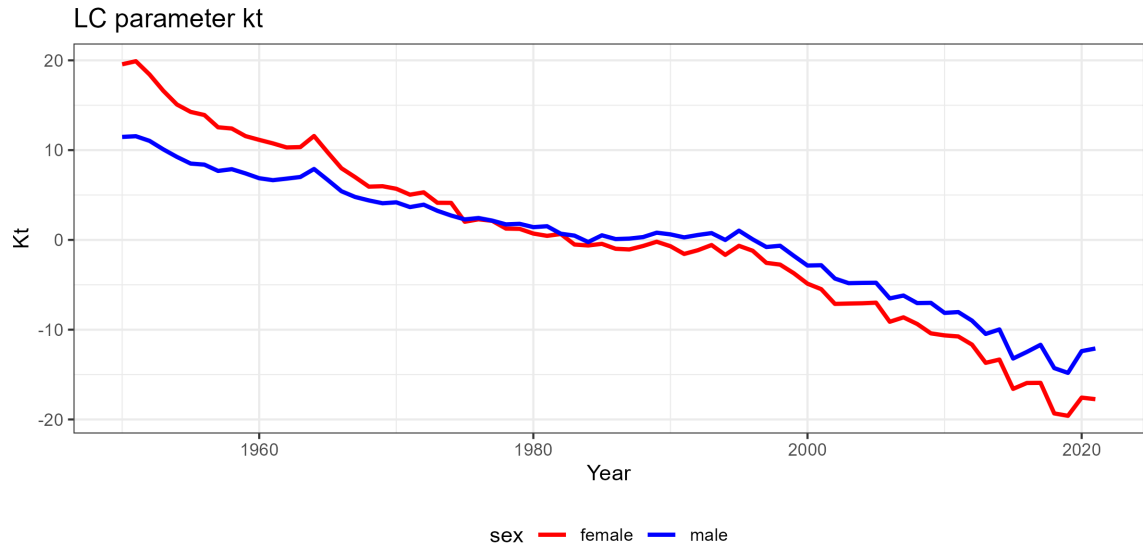


Here we can see the sensitivity of the mortality index to changes in mortality. The highest sensitivity is at early life. This is due to the monumental work done on the island in the latter half of the 20th century, better sanitation, better nutrition, and better healthcare. The high point is followed by a significant drop for both sexes at the teen years. Women then transition to a steady decline in sensitivity but males continue to drop until near zero at the age of 20 finally making a slow

climb through to meet the female sensitivity at around 80 years of age. This is a clear example of the context specific nature of the bx parameter, as the sensitivity of mortality to changes in the overall trend is not uniform across age groups and can be influenced by various factors such as healthcare improvements, lifestyle changes, and social determinants of health that affect different age groups differently. We will specifically see the effect of the parameter in the death rates and life expectancy projections later on, but for now we can move on to the kt parameter, which is our time-varying mortality index.

Fitted kt

The last of the initial fitted parameters, kt serves as the time-varying mortality index that captures the overall trend in mortality improvement over time. Historically, the kt parameter typically shows a declining trend over time, reflecting improvements in healthcare, living conditions, and other factors that contribute to reduced mortality rates. The kt parameter can also exhibit fluctuations due to various events such as epidemics, wars, or economic downturns that can temporarily increase mortality rates. For our case, the kt parameter graphs as follows.



The index corresponds to the average level of mortality for that year. There are fluctuations in the index, a spike in the mid 1960s, the effect of HIV in the mid 1990s (Calzada & Torres, n.d.) but the overall trend is a steady decline. The index starts off higher for women than men but they close the gap by the mid 1970 and start pulling apart as time passes leading to an slow growing gap, but both share the same general shape with the male and female indexes crossing in the mid 1970s.

Parameter simulation

Now that we have the initial fit, we can move on to the simulations. We provide the fitted parameters to the bootstrap method, which resamples the parameters to create new sets of parameters. These new parameters come with their own residuals to evaluate which we use to determine the number of bootstraps needed to create a robust distribution. Earlier, we stated that 500 samples were sufficient and these are the corresponding residual graphs and abridged tables of median

and quantiles. The unabridged ones will be at the end in the Appendix.

ax residuals

Below we have the *ax* residuals of the bootstrap. It presents a large deviance spike around childhood. For males this tapers quickly when entering the teen years and continues to narrow through adulthood and the senior years. Women on the other hand present a slower narrowing into adulthood and further slows down past the 40 year mark. Looking at the quantiles, at the median there's a clear skew towards opposite sides of zero between the sexes and the range between the 95% quantiles between men and woman are within 20% of each other. In short both share the same general shape of uncertainty with the males having a slight offset towards the negative.

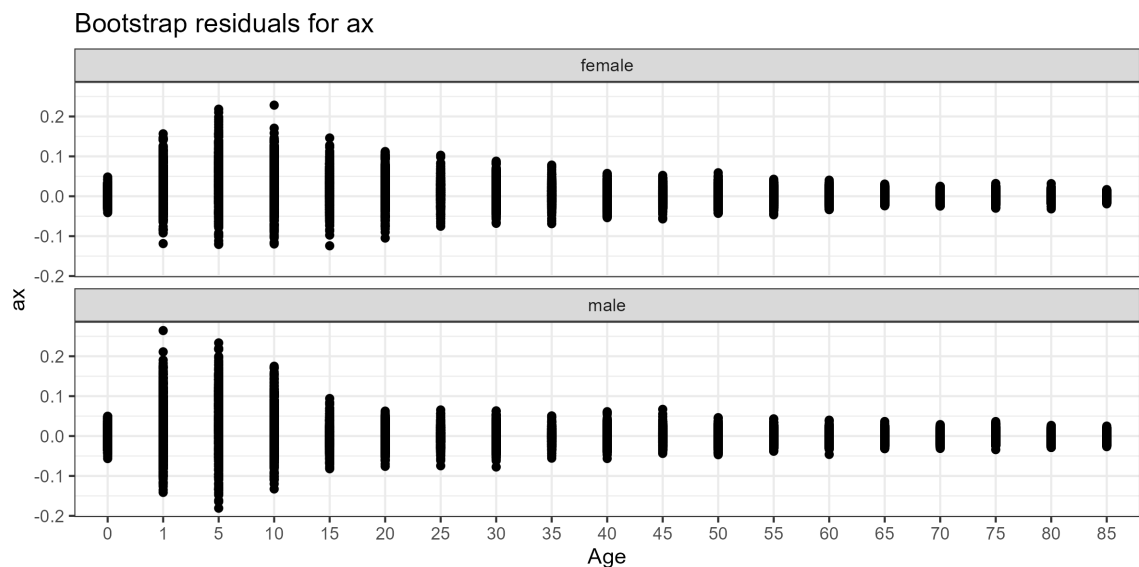


Table 1: Bootstrap residuals for ax

Age	Male	Female
	Median (2.5%, 97.5%)	Median (2.5%, 97.5%)
0	-0.000488 (-0.035552, 0.038609)	0.003181 (-0.028166, 0.030445)
10	0.018548 (-0.080141, 0.129351)	0.03457 (-0.07385, 0.131551)
20	-0.002768 (-0.043945, 0.042574)	0.013768 (-0.05207, 0.080922)
30	-0.000139 (-0.041437, 0.039752)	0.008157 (-0.040066, 0.062687)
40	-0.000751 (-0.035567, 0.032203)	0.00358 (-0.037061, 0.041453)
50	-0.000515 (-0.028436, 0.026259)	0.00207 (-0.030501, 0.033416)
60	-0.002996 (-0.025405, 0.021127)	0.002229 (-0.02133, 0.024467)
70	-0.001699 (-0.020002, 0.018293)	0.000511 (-0.018162, 0.018881)
80	-0.002047 (-0.023727, 0.018088)	0.000993 (-0.016665, 0.017494)

bx residuals

Moving on to the bx parameters bootstrap residuals, in many ways it mirrors the ones from the ax parameter just an order of magnitude lower, with a large deviance spike around childhood that tapers off as age progresses. However, unlike the ax residuals, we see a clear difference in the range of the residuals between sexes. For males, the residuals are much more spread out than the female counterpart, with the 95% range generally being wider for men more often than in the ax residuals and by a larger margin. This leads us to derive that the male sensitivity to changes in the mortality index is much more volatile than the female sensitivity.

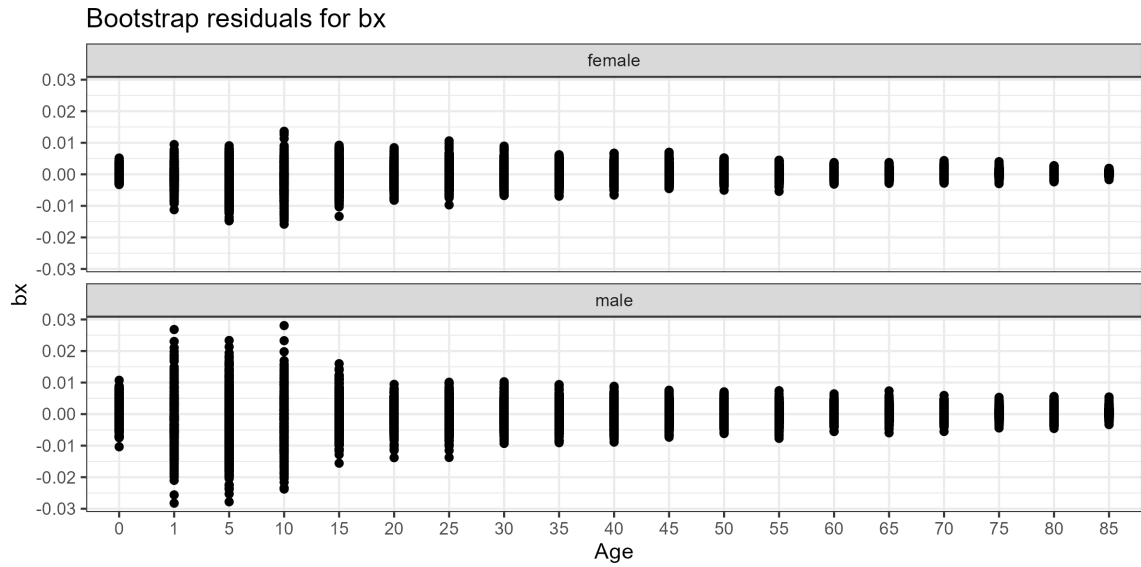


Table 2: Bootstrap residuals for bx

	Male	Female
Age	Median (2.5%, 97.5%)	Median (2.5%, 97.5%)
0	0.00114 (-0.005187, 0.007502)	0.000345 (-0.002173, 0.003594)
10	-0.001605 (-0.018327, 0.013581)	-0.001329 (-0.011613, 0.007072)
20	0.000299 (-0.007026, 0.006821)	-0.000358 (-0.006262, 0.005847)
30	0.000273 (-0.005946, 0.006914)	9.8e-05 (-0.005167, 0.005623)
40	0.000395 (-0.006065, 0.006222)	0.000417 (-0.003907, 0.004487)
50	0.000399 (-0.003973, 0.005129)	0.000419 (-0.002729, 0.003733)
60	0.000497 (-0.002977, 0.004151)	0.000425 (-0.00184, 0.002889)
70	0.000572 (-0.002522, 0.004003)	0.000345 (-0.001594, 0.002613)
80	0.000466 (-0.002924, 0.003582)	0.000426 (-0.001337, 0.002105)

kt residuals

Moving on to the residuals for the *kt* variable, we see that unlike both other parameters, *kt* slowly fans out over time. The range is much larger than the other two parameters, which is expected as the *kt* variable is the one that is projected forward in time and thus has more uncertainty associated with it. The medians trend upwards over time and the range spread is similar on both sexes. Generalizing, there is a clear increase in uncertainty as time progresses.

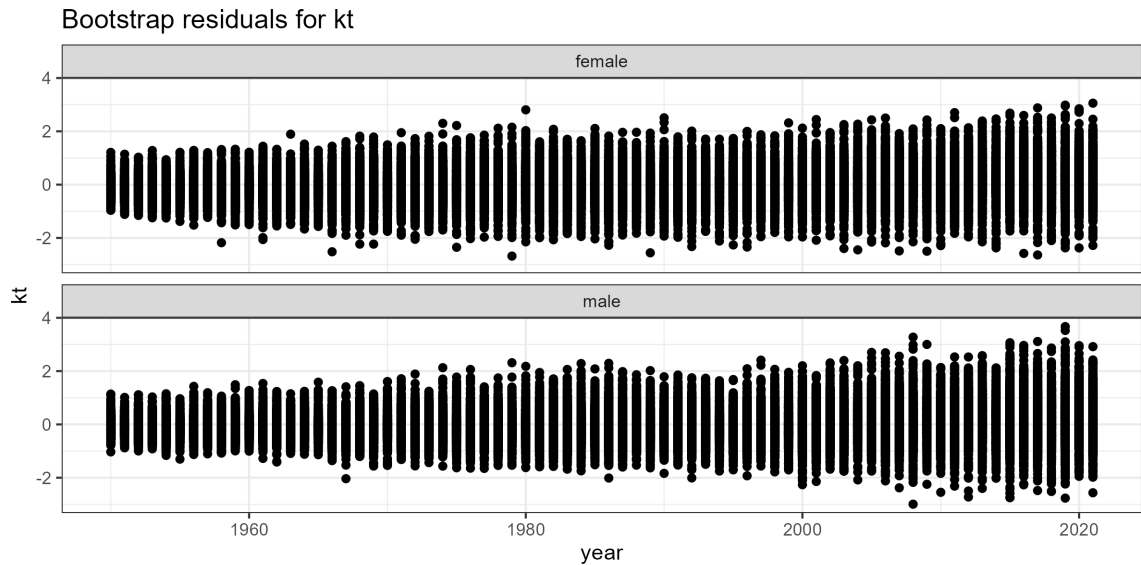


Table 3: Bootstrap residuals for *kt*

Male		Female
Year	Median (2.5%, 97.5%)	Median (2.5%, 97.5%)
1950	-0.11087 (-0.655904, 0.564723)	-0.114965 (-0.761783, 0.634526)
1960	-0.084436 (-0.777783, 0.810472)	-0.102494 (-1.149363, 0.709864)

1970	-0.069452 (-0.968262, 0.932076)	-0.110846 (-1.313768, 1.046471)
1980	-0.082886 (-1.067248, 1.043432)	-0.027 (-1.278711, 1.287512)
1990	-0.080615 (-1.092941, 1.09753)	-0.04279 (-1.415906, 1.200716)
2000	0.001651 (-1.293733, 1.589222)	0.118993 (-1.300456, 1.314144)
2010	0.087276 (-1.300719, 1.632667)	0.09324 (-1.360864, 1.537258)
2020	0.242392 (-1.453318, 2.113448)	0.354226 (-1.193514, 1.657209)

The forecast

Now, that we have the bootstrap parameters, we can now do the simulations. The simulations are done by taking the bootstrapped parameters and simulating the kt variable forward in time. This allows us to create a distribution of possible future mortality scenarios based on the uncertainty in the parameters. The results of the simulations can be used to create confidence intervals for the future mortality rates. The parameters that define the simulations are the number of simulations, the years to project, the jump choice, and the method of projection for kt . For our case, we set the number of simulations to 500, the same as the bootstrap samples, the years to project, 40 in our case, the jump choice, the actual end in our data. The method of projection for kt , either multivariate random walk with drift or $iarima$ models, the vast majority of literature uses the random walk and we have no reason to deviate from this (Girosi & King, 2007).

Simulated kt

Now, looking at the projections for the kt variable, we see that both medians fall at a steady rate over time. As expected of the random walk both ranges spread out over time with them both presenting a 10 point differential in at the first projection point finally growing to a 30 point differential by the end of the projection period. Of note is the skew that the median projections have towards the lower bound indicating a more optimistic projection for the future mortality trends.

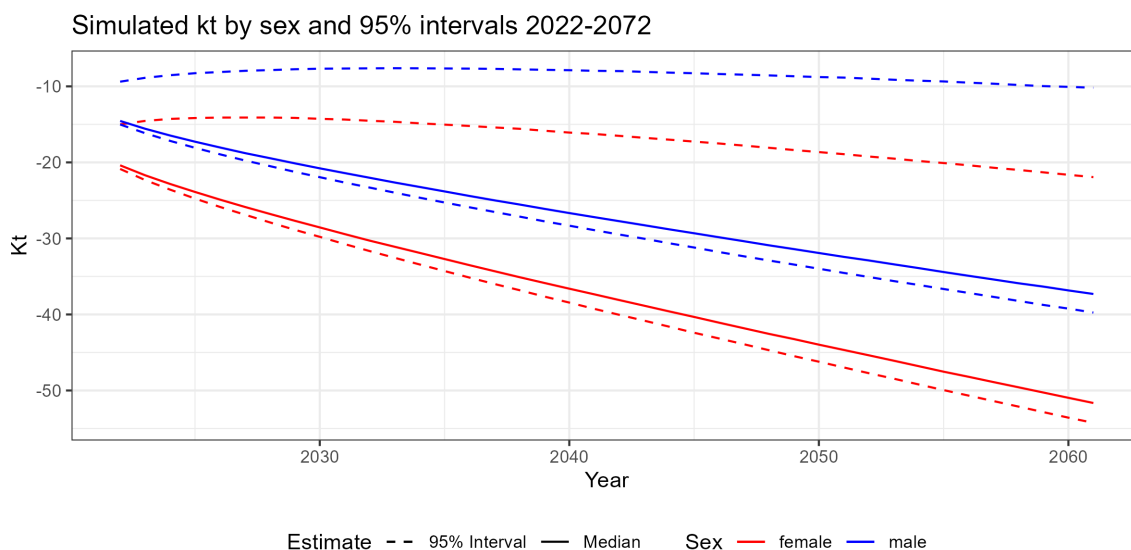


Table 4: kt simulations 2022-2050.

	Male	Female
Year	Median (2.5%, 97.5%)	Median (2.5%, 97.5%)
2025	-17.277059 (-18.081038, -8.269409)	-23.88656 (-24.755169, -14.163912)
2030	-20.781023 (-21.946493, -7.676944)	-28.552782 (-29.791317, -14.266972)

2035	-23.818407 (-25.276301, -7.646628)	-32.709276 (-34.288534, -15.033035)
2040	-26.66492 (-28.325247, -7.873816)	-36.611439 (-38.438835, -16.077728)
2045	-29.319704 (-31.194549, -8.274248)	-40.326325 (-42.404781, -17.265159)
2050	-31.916872 (-33.98382, -8.776932)	-43.972697 (-46.213298, -18.644115)
2055	-34.421887 (-36.653311, -9.352436)	-47.52888 (-49.950673, -20.086872)
2060	-36.8401 (-39.230637, -10.066605)	-50.963188 (-53.575821, -21.630627)

Results

Death rates

Now, with the kt projections, we can now calculate the first of our two final products, the projected mortality rates for each age and sex by using the ax and bx parameters. The projected mortality rates can be calculated using the formula: $\log(mortalityrate) = ax + bx * kt$. This allows us to create a distribution of possible future mortality rates based on the uncertainty in the parameters and the projections of kt . The projected mortality rates can be used to create confidence intervals for the future mortality rates.

Table 5: Projected log mortality rates for selected ages and years.

Year	Age	Male Log Death Rate (95% CI)	Female Log Death Rate (95% CI)
2000	20	-6.0309 (NA, NA)	-7.7218 (NA, NA)
2000	40	-5.402 (NA, NA)	-6.3572 (NA, NA)
2000	60	-4.1723 (NA, NA)	-4.8466 (NA, NA)
2025	20	-6.1843 (-6.1653, -6.2142)	-8.0819 (-7.7618, -8.4002)
2025	40	-5.3752 (-5.2547, -5.4991)	-6.6737 (-6.4647, -6.8816)
2025	60	-4.3576 (-4.2353, -4.4804)	-5.1072 (-4.9603, -5.2531)
2050	20	-6.2013 (-6.1298, -6.3123)	-8.8978 (-8.04, -9.7656)
2050	40	-5.5881 (-5.2655, -5.9312)	-7.2056 (-6.6468, -7.7747)
2050	60	-4.5752 (-4.2456, -4.9104)	-5.4815 (-5.0878, -5.8788)



Here we have one of our final products, the death rates. Using the median projection and dividing by age group and sex, we see that minors in both sets are projected to have a continued accelerated mortality decline through the projected period though their mortality is highly erratic especially by the end of our historic data. Adult women are also projected to have a continued decline in mortality but at a slower pace than minors, while mortality is projected to plateau for young adult. Those that survive to late adulthood are projected to continue their decline in mortality but at a much slower pace than that of women. Finally for both senior groups we see a continued decline in mortality with women projected to have a much faster decline. One of the more important things to note is that the original bx parameter describes much of this graph. The high sensitivity of the minors to changes in the mortality index kt means that they are projected to have a much faster decline in mortality than the other age groups. The low sensitivity of the young adults

means that their mortality is projected to plateau, while the moderate sensitivity of the adults and seniors means that they are projected to have a continued decline in mortality but at a slower pace than minors. Meanwhile the ax parameter is over-written by our explicitly requirement to force the projections to start at the actual end of the data, so it does not have a direct effect on the projections but it does have an indirect effect by influencing the initial fit of the model and therefore the parameters that are used in the simulations.

Life Tables and Life expectancy at birth

Now for the second final product. We produced life tables for each of the projected mortality rates, which allows us to calculate life expectancy at birth for each of the projections. This gives us a distribution of possible future life expectancy at birth values based on the uncertainty in the parameters and the projections of kt . The following abridged life tables refer to the last year of the median projection and the following graph shows the life expectancy at birth projections with confidence intervals.

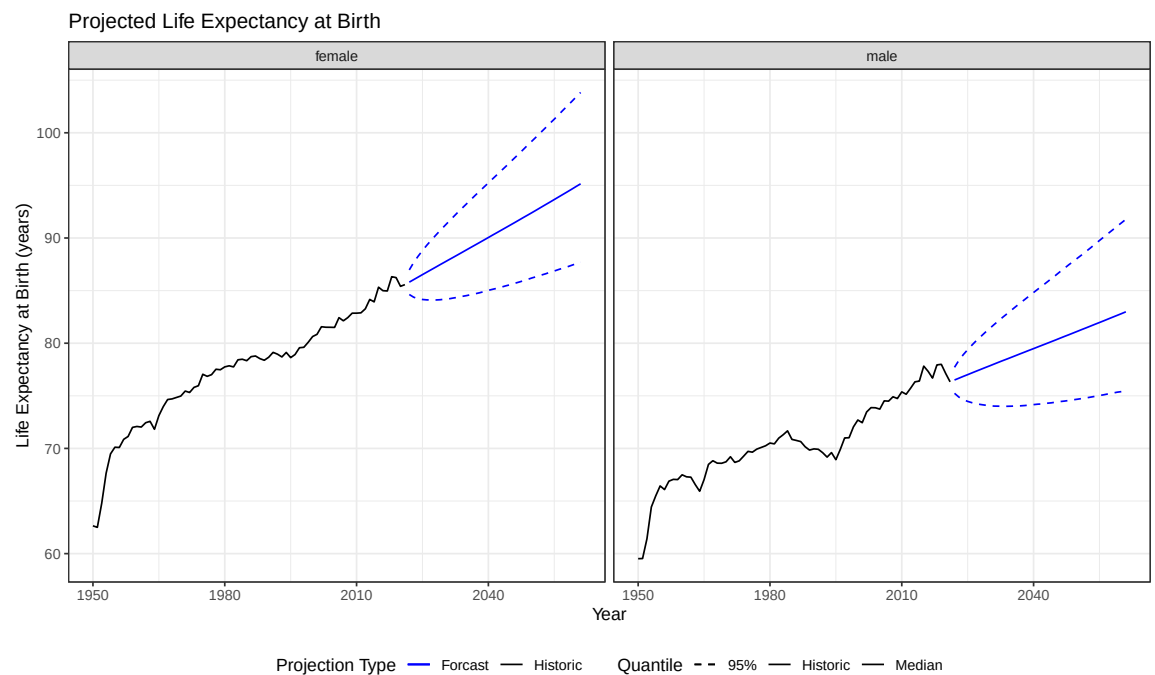
Table 6: Life Table Male 2061.

Age	ex	qx	lx	dx	Lx	Tx
0	82	0.0014	100000	137	99869	8298399
20	63	0.0100	99568	995	495352	6302074
40	45	0.0169	94971	1605	470842	4353076
60	29	0.0457	85851	3928	419435	2529381

80	16	0.1730	62323	10783	284657	1009021
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Table 7: Life Table Female 2061.

Age	ex	qx	lx	dx	Lx	Tx
0	95	0.0013	100000	127	99879	9515069
20	75	0.0005	99841	47	499087	7517787
40	55	0.0029	99392	291	496232	5524105
60	36	0.0175	96632	1691	478932	3556420
80	20	0.1055	83327	8794	394650	1722172



Seeing the life expectancy at birth projections, we see that the historic data shows a steady increase in life expectancy for both sexes and a dip around the 1990s for men. The projected values follow this trend with a continued increase in life expectancy of about 2.3 years per decade for women and 1.6 years per decade for men. The median of the projection reaches 94 years for women and 82 for men.

The confidence intervals on the other hand present a curious picture. The projection intervals are large enough to warrant a closer look, in particular, the upper bound of the female life expectancy projection presents linear growth and is projected to reach 104 years by 2060 and male life expectancy reaches 92 year. The lower bound for both seem much more plausible, with the female projection reaching 89 years and the male 76.

Discussion

Our benchmarks

We have already stated that there are no modern projections of mortality and life expectancy projections that are specific to Puerto Rico, so we will be comparing our results to three entities that do produce projections for the entire world, including Puerto Rico: the UN World Population Prospects (WPP), the U.S. Census Bureau's International Data Base (IDB), and the University of Washington's Institute for Health Metrics and Evaluation (IHME). Each of these entities uses different methodologies and assumptions, so comparing our results to theirs will provide a useful context for understanding the implications of our projections.

The WPP projects mortality using a two-phase probabilistic model that combines historical trends with country-specific data and model life tables. It begins by projecting life expectancy at birth using a double-logistic function, capturing rapid improvements in early mortality followed by slower gains in older ages. For data-rich countries, the Lee-Carter method is applied to age-specific mortality rates, while for data-scarce countries, model life tables (e.g., Coale-Demeny or UN families) are calibrated using indicators like under five mortality and adjusted via the Calibrated Spline method. Regional benchmarks temper extreme trends such as rapid gains in fragile states or stagnation in advanced nations ensuring plausible long-

term projections. The WPP also incorporates the impact of crises like pandemics and conflicts through excess mortality estimates, and uncertainty is quantified using prediction intervals. These mortality projections are then used in cohort-component models to forecast population change.

The IDB produces mortality and life expectancy projections for Puerto Rico and other global regions using a multi-step cohort-component methodology. Base mortality estimates are derived from high-quality data sources like vital registration systems, surveys and censuses, with adjustments for underenumeration, age heaping, and data quality issues. For countries with limited data, model life tables are used to estimate age and sex-specific mortality patterns. The IDB extends mortality projections to age 100+ using logistic functions to ensure smooth, realistic trends, even in the absence of empirical data. For countries affected by HIV/AIDS, the IDB employs a dual-scenario approach, modeling mortality under both “with AIDS” and “without AIDS” conditions using data from the UNAIDS Spectrum software to incorporate HIV prevalence and epidemic dynamics. Projections for future mortality are based on logistic curves fitted to historical life expectancy trends, with fixed slope models ensuring consistent long-term improvements, capped at 95 years for males and 100 for females by 2100. These projections are validated against demographic indicators and adjusted for shocks like wars, famines, and natural disasters, ensuring high precision in capturing the timing and impact of major events.

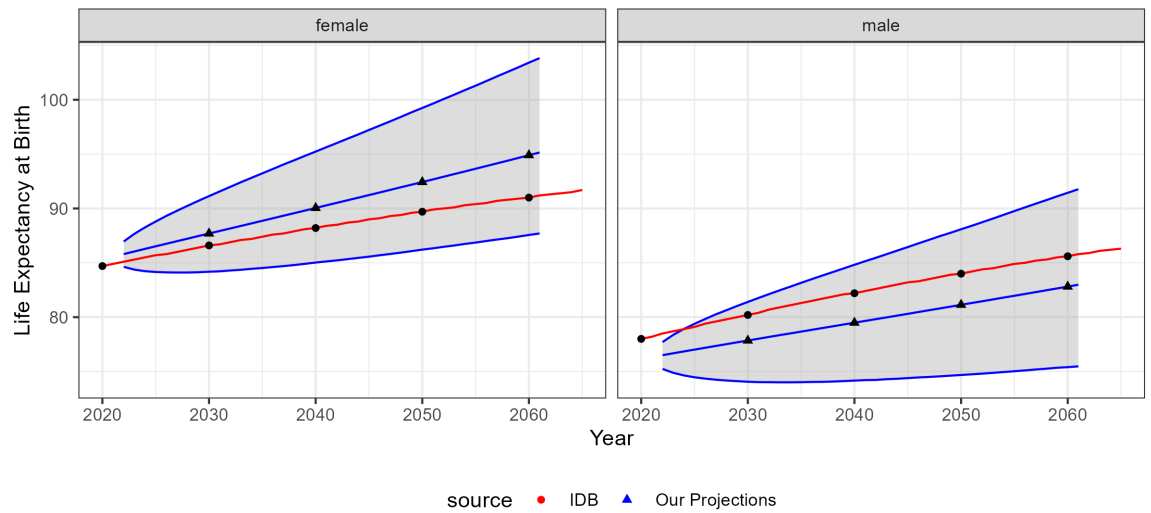
Finally, the IHME projects mortality using a data driven model that integrates the Socio-demographic Index, cause-specific risk factors from the Global Burden of Diseases study, and an ARIMA component to capture unexplained mortality trends.

The model estimates underlying mortality as a function of SDI, time, and risk factors, while explicitly accounting for the interplay of risk factors such as tobacco, obesity, and air pollution. An ARIMA model with attenuated drift is used to project residual mortality, ensuring robust long-term forecasts without overfitting. This approach enables IHME to generate probabilistic, scenario-based projections for mortality and population, incorporating uncertainty across all components and providing a dynamic, causal framework distinct from more deterministic models used by other organizations.

Comparison with External Benchmarks

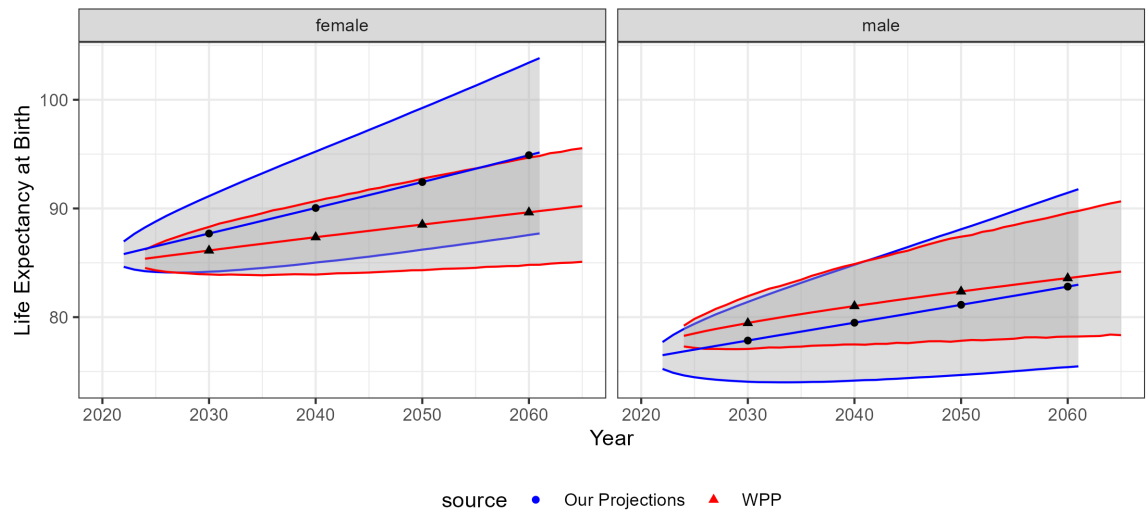
With our actors and their methodologies defined, we can now compare our projections to theirs. The nature of the International Databases projections mean that they do not produce confidence intervals, so we will be comparing our median projections to their point estimates. In the graph below, we can see that our median projections fall on opposite ends of the IDB. Our female mortality projection starts slightly above the IDB's point estimate and then diverges upwards slowly, while our male mortality starts below the IDB point estimate point and keeps the gap until the end of the projection period. In short, females will outpace the IDB projections while the males will consistently lag behind the IDB projections. In both cases, the IDB projections lie within our confidence intervals.

Comparison of our projections with the IDB: 2023:2062



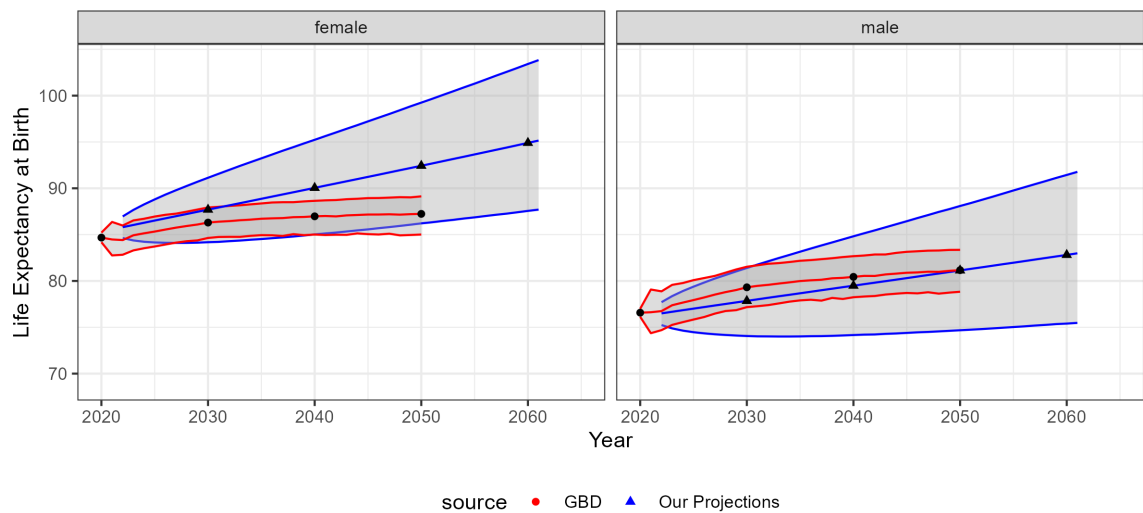
With the WPP, we can see a different pattern. In our case, our median line for female life expectancy starts above the WPP median but instead of tracking the WPP median, it diverges upwards tracking the WPP upper bound, while our female upper bound diverges even more upwards. For males, the story is more plausible, with our male median starting below the WPP median but climbs faster than the WPP median. The upper bound of our male projections tracks the WPP upper bound until the second half of the projection period where it diverges upwards, while the lower bound is always below the WPP lower bound. This is a more plausible pattern.

Comparison of our projections with the WPP: 2023:2062



Finally, the IHME projections are much more difficult to compare to our projections. They do provide confidence intervals, but they don't produce the typical fanning out of the confidence intervals that we would expect from a stochastic model. Instead, their confidence intervals are much more narrow and they don't show the same pattern of divergence that we see in our projections. Their model implies that there will be a plateauing of life expectancy for both males and females in the 2040s, while every other projection, including ours, shows a continued increase in life expectancy through the end of the projection period. This is a significant divergence from our projections and the other benchmarks. That being said, the IHME projections generally lie within our confidence intervals, though our medians behave very differently from the IHME medians, with our medians showing a continued increase in life expectancy while the IHME medians show a plateauing of life expectancy. This is a significant divergence from our projections and the other benchmarks.

Comparison of our projections with the GBD: 2023:2062



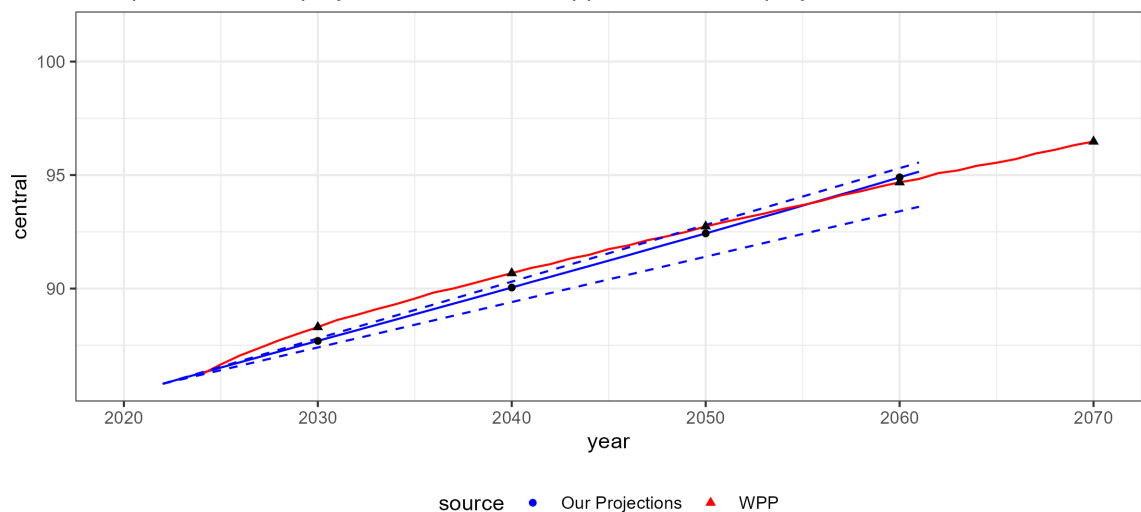
When we see the then median of the male projections for the WPP and IDB, we see that they are quite close to our median projections, with our projections for 2061 running just below the WPP at 82 years the WPP at 83 years and the IDB at 85 years. Our intervals do behave differently with ours presenting much more uncertainty in the lower bound growing much faster than the WPP then curving back up, until the end of the projection period where it stops at 75 years while the WPP's lower bound reaches 78 years. The upper bound of our projections roughly tracks the WPP until it starts diverging upwards after 2045, reaching 91 years by 2061 while the WPP's upper bound reaches 89 years.

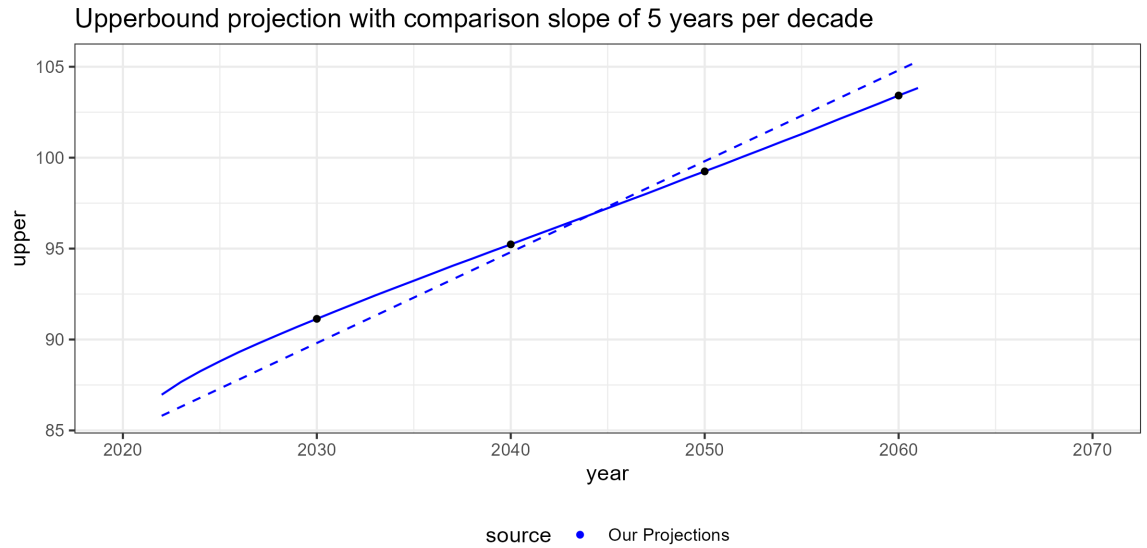
The male projections have a clear downwards drift on the lower bound at the beginning of the projection period, but it presets a plausible trajectory given the much more erratic nature of male mortality i.e. its not shocking. What is shocking is the behavior of the female projections, where the median projection passes the upper bound of the WPP projections by the end of the projection period with

ours reaching 94 years while the WPP reaches 90 years. The lower bound of our projections reaching 95 years with the IDB presenting 91 and the WPP 89. The upperbound for our projections reaches 103 years while the WPP reaches 94 years. and the lower bound of our projections reaches 87 years while the WPP reaches 84 years.

Oeppen and Vaubel stated in May 2002 that “best-practice life expectancy gains” have been about 2.5 years per decade(Oeppen & Vaupel, 2002), meanwhile Ronald Lee states its likely closer to 2.0 years per decade (R. Lee, 2002). If we graph those limits, we can see that our median projection for women exceeds the 2.0 year per decade limit stated by Ronald Lee, and approaches the 2.5 year per decade limit stated by Oeppen and Vaubel. This isn’t even considering the upper bound of our projections, which exceeds the 2.5 year per decade limit by a wide margin, reaching 5.0 years per decade as we show below

Comparison of our projections with WPP upper bound and projection median with theoretical





The Skewing of female mortality

Its well known that the Lee-Carter model can produce skewed projections for life expectancy, particularly for longer projection horizons, due to the way uncertainty in the k_t index translates into life expectancy. In our projections for Puerto Rico, this skew is particularly pronounced and we have done some analysis to understand why that is the case. The usual suspects have been ruled out, as the model fit is good, the b_x parameters are reasonable and the historical k_t index does not show any unusual patterns that would suggest a structural break. Following those checks weve double checked the code, both verifying the implementation of the Lee-Carter model and the random walk with drift projection of k_t , we have iterated through all the parameters to see if any of them could be driving the skew. In fact, the the parameters themselves do little to change the structural shape of the projections, with both the median and the upperbound of female mortality showing

unrealistic improvements in life expectancy at birth, while the lower bound shows a slightly more plausible trajectory under all scenarios.

Model Performance and Limitations

Pleateau in young males

It is well known that the young adult male population in Puerto Rico has been affected by high rates of violence and accidents, which has led to excess mortality in that age group. The Lee-Carter model captures this historical pattern through the b_x parameters, which shows a very low sensitivity to changes in the k_t . Most of the enviromental change over the past decades have been in areas of healthcare and public health. It is likely that these changes have been canceled out by the excess mortality from violence and accidents, leading to a plateau in the projections for young males. But this is a known limitation of the Lee-Carter model, which does not account for changes in the underlying causes of mortality or for structural breaks in the data. The model assumes that the age-specific mortality rates will continue to evolve in a similar way to the past, which may not be the case for Puerto Rico given the unique challenges it faces.

Future shocks

The model was able to properly capture the mortality shock of the mid 90s from HIV. However, the random walk with drift projection of the k_t index does not

allow for the anticipation of future shocks, which is a known limitation of the Lee-Carter model. This is particularly relevant for Puerto Rico, which has a history of hurricane-driven excess mortality. The model may not be able to capture the potential impact of future hurricanes or other natural disasters on mortality rates, which could lead to an overestimation of life expectancy in the long run.

Broader Implications

Even if our projections outperform the benchmarks by an unrealistic margin, the data tells us two things. First, that there is a significant amount of uncertainty in the projections, particularly for females and particularly in the upper limits of mortality, which is reflected in the wide confidence intervals. This uncertainty is not just a modeling limitation but a reflection of the complex and dynamic nature of mortality trends in Puerto Rico, which are influenced by a wide range of factors including healthcare, public health, social and economic conditions, environmental and migratory factors. Second, that life expectancy in Puerto Rico is likely to continue to improve in the future, but the rate of improvement is uncertain and may be influenced by a wide range of factors that are difficult to predict. This has important implications for policymakers and planners, who need to be prepared for a range of possible scenarios when it comes to healthcare and social services for the aging population in Puerto Rico. The wide confidence intervals themselves are a finding, they communicate to policymakers that demographic uncertainty in Puerto Rico is not merely a modeling limitation but a genuine planning risk that demands

scenario-based fiscal and healthcare planning rather than single-point forecasts.

Further more a deeper look into the source of the skew is needed to understand the underlying drivers of mortality trends in Puerto Rico and to improve the robustness of our projections. The two main data sets, local census estimates for population and the local health department for vital statistics, are the likely sources of the skew. The census might be overestimating the population on the latest decennial census which was done during the pandemic. But this would mean there is a significant over counting of the population, leading to a much lower mortality rate over the past decade and the years that follow. There have been no vital statistics reports in the past years, but the health department reports to the national vital statistics system, so it is unlikely that there has been a significant under counting of deaths since we were not filtering for any specific cause of death. Lastly the boogiemans in demography in Puerto Rico is the out migration over the past decade. But for this to be the cause it requires that the population leaving specifically have a higher likelihood of dying than the population that remains, which is possible but we have no evidence to support nor believe this claim. Further research is needed to understand the source of the skew and to improve the robustness of our projections, but it is important to note that the skew is not a modeling artifact but rather a reflection of the data and the events that have shaped mortality in Puerto Rico in recent years.

Recomendations

The underlying model we used, the Lee-Carter model, is a widely used and well-established model for projecting mortality and life expectancy. Having said that, it's clear that the model has limitations, particularly in the context of Puerto Rico where there are unique challenges and structural breaks in the data. As such our recommendation is to utilize the hierarchical Bayesian model developed by the United Nations World Population Prospects (Raftery et al., 2013), which incorporates a much larger context of data and events, and is likely to be more robust to the kinds of structural breaks that we have identified in our analysis. Having said that, the Lee-Carter model and this analysis still useful. The hierarchical Bayesian model is a much more complex model that requires a much larger amount of data and computational resources. While it does generate much more believable projections, one would not be able to identify the structural breaks in the data that we have identified with the Lee-Carter model, which is a useful finding in and of itself. The Lee-Carter model is also much more transparent and easier to understand and interpret, which can be useful for communicating the results to policymakers and stakeholders.

Future work

The next step in this project is to implement the Hierarchical Bayesian Model developed by the United Nations World Population Prospects (Raftery et al., 2013) and compare its projections to those generated by the Lee-Carter model. Additionally, the source of the skew in the projections needs to be further investigated, particularly in relation to the data from 2010 to 2023. This could involve a deeper analysis of the events that occurred during that period and how they may have impacted mortality trends in Puerto Rico, as well as an examination of the data collection methods and quality during that time. Finally, it would be useful to explore other modeling approaches that can account for structural breaks and changes in the underlying causes of mortality. This could include models that incorporate covariates related to healthcare, public health, social and economic conditions, environmental and migratory factors as we have not included any covariates in our projections. Finally, these projections should be used to inform scenario-based planning for healthcare and social services for the aging population in Puerto Rico, taking into account the wide range of possible outcomes and the significant uncertainty in the projections as well as serve as a stand-in for the more complex hierarchical Bayesian model until we are able to implement it.

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